

EXPERIMENT 31: MORPHOLOGICAL OPERATIONS USING OPENCV – OPENING

Aim

To perform the morphological Opening operation on an image using Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Convert the grayscale image into a binary image using thresholding.
5. Create a structuring element (kernel).
6. Apply the Opening operation using `cv2.morphologyEx()`.
7. Display and save the opened image.

Python Code

```
import cv2
import numpy as np

# Read the input image
img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Convert image to binary
_, binary = cv2.threshold(
    gray, 120, 255, cv2.THRESH_BINARY
)
```

```
# Create structuring element
kernel = np.ones((5, 5), np.uint8)

# Apply Opening operation
opening = cv2.morphologyEx(
    binary,
    cv2.MORPH_OPEN,
    kernel
)

# Save output
cv2.imwrite("opening_output.jpg", opening)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Binary Image", binary)
cv2.imshow("Opening Image", opening)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the morphological Opening operation was successfully performed on the input image using OpenCV.